Auto-Decte: Trust-Constrained Human-in-the-Loop Document Intelligence for Industrial Paper Forms

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Abstract. High-stakes document extraction — such as digitizing paper payroll and production forms — demands more than accuracy: a single mis-transcribed digit changes a worker's wage, so machine errors must never silently enter the factual record. We propose a trust-constrained human-in-the-loop framework for document intelligence, instantiated in Auto-Decte, a deployed system for industrial paper forms. The framework rests on two principles: (i) *candidates, never facts* — every machine output (perception, LLM suggestion, or retrieval result) is an append-only candidate or a read-only reference, and only human confirmation creates a versioned fact; and (ii) *abstention over guessing* — recognition emits an explicit ambiguity band and abstains on borderline cells, routing them to review instead of risking silent errors. Perception is fiducial-guided (QR classification, ArUco perspective correction) rather than learned; a DeepSeek adapter and a local retrieval index assist reviewers under the same trust boundary; a PWA with an offline outbox extends submission to the plant floor. We evaluate recognition accuracy, a selective-prediction coverage-vs-risk curve, robustness under perturbation, retrieval precision, end-to-end whole-form accuracy (97.5% clean), and traceability/AI-off invariants (389 automated acceptance tests).

Keywords: trustworthy document intelligence · selective prediction · human-in-the-loop · information extraction · LLM/RAG assistance

1 Introduction

Foundation-model-assisted document processing is attractive for the long tail of paper-based workflows in small and medium plants, but payroll-adjacent settings impose a stricter requirement than raw accuracy: trust. An LLM that transcribes one digit wrong changes a worker's wage; a retrieval system that injects a similar-looking historical record can silently corrupt a decision. Fully automatic pipelines are unacceptable where errors carry financial consequences, yet purely manual digitization does not scale. This motivates our central research question: *how can machine learning and generative assistance extract information from high-risk paper documents while guaranteeing that machine errors never enter the factual data layer?*

We address this with a trust-constrained framework whose two principles are (i) *candidates, never facts*, and (ii) *abstention over guessing*. We study three sub-problems: (RQ1) reducing perception uncertainty on structured paper forms (template-guided perception with an explicit ambiguity band); (RQ2) letting LLM/RAG provide assistance without contaminating facts (candidate/reference isolation); (RQ3) making human confirmation the sole path that writes facts while preserving traceability (append-only candidates, versioned decisions, evidence lineage).

Our contributions are:

1. A *trust-constrained document intelligence framework* separating an evidence layer, a candidate plane, a human trust boundary, and a fact plane, enforcing the invariant that machine outputs never become facts.
2. *Selective recognition via abstention*: a per-cell ambiguity band (template-matching margin for digits, fill-ratio band for OMR) that abstains on borderline cells; we report the coverage-vs-selective-risk trade-off.
3. *Trustworthy LLM/RAG integration*: suggestions are stored separately and cannot mutate confirmed values, retrieval is strictly read-only, and the AI-off path keeps the pipeline correct.
4. An offline-capable mobile submission path and a deployment case study with end-to-end acceptance.

2 Related Work

Document understanding. General document AI stacks (PP-OCRv3 [1], LayoutLMv3 [2], Donut [3]) and OMR systems (SurveyNet [4]) learn to parse free-form documents, but they treat machine output as the answer rather than as a candidate to be confirmed. **Fiducial localization.** ArUco detection [5] and MILP marker dictionaries [6] provide the geometric anchoring we adopt for deterministic template recovery. **Human-in-the-loop.** Active learning and HITL surveys [7] motivate human review, but typically to improve model accuracy, not as the sole write path to authoritative records. **LLM/RAG assistance.** Generative information extraction and RAG surveys [8, 9] and local-first principles [10] inform our optional assistant and offline submission. Individually, none of these components is new; our contribution is their integration into a single enforced architecture in which selective abstention and a machine-to-fact isolation boundary operate together, a combination we are not aware of in prior document-intelligence systems.

3 Problem Formulation and Design Principles

High-risk document extraction. A paper form is photographed and mapped to structured field values $y = \{y_j\}$. The requirement is not only that a prediction $\hat{y}_j$ be accurate, but that any machine error be *detectable and non-persistent*: a wrong machine output must not survive into the stored record without human awareness.

Candidate vs fact. We distinguish three data planes — evidence, candidate, and fact — separated by a human trust boundary. The *evidence plane* holds immutable, content-addressed inputs (SHA-256 images). The *candidate plane* holds everything a machine produces — recognition hypotheses, LLM suggestions, retrieval references — all append-only and forbidden from overwriting facts. The *fact plane* holds versioned values that exist only after a human confirms or corrects them. This separation is architectural and enforced by the data model.

Abstention over guessing. A recognizer should predict only when its uncertainty is low, otherwise abstain and route the cell to review. We operationalize this with a per-cell confidence and an explicit ambiguity band, making abstention a first-class output rather than a post-hoc threshold.

Trust model. Let E, C, H, F denote the evidence, candidate, human-decision, and fact planes. Every machine module $m \in M$ (perception, LLM, retrieval) may write only into C; the fact plane evolves solely through human decisions, $F_{t+1} = g(F_t, H_t)$. The framework enforces the invariant $\forall m \in M : m \not\to F$ — no machine output has a direct write path to the fact plane. This invariant is *architectural* (enforced by the data model) and guarantees that a machine error can never silently become a fact; it is distinct from any empirical claim about recognition error rate, evaluated separately in Section 6.

4 Trust-Constrained Document Intelligence

Template-guided perception. Templates are generated with a QR code (identity + version) and ArUco corner markers (IDs 10–13). A multi-region QR search classifies the form; four ArUco markers map it to a canonical canvas via perspective correction; fields are cropped from the known template definition. Identification and geometry are thus deterministic and auditable, confining recognition to per-cell decisions.

Selective recognition with an ambiguity band. Digit cells are normalized to 48×64, binarized (Otsu), and matched against ten rendered synthetic-font templates by normalized mean absolute difference. The recognizer reports the top digit and abstains (flags AMBIGUOUS) when the margin to the second-best template is below a threshold τ. Blank cells (ink ratio $< 1\%$) are reported as blank. Checkbox (OMR) cells use an interior fill ratio: ≤ 0.15 unchecked, ≥ 0.35 checked, and the $(0.15, 0.35)$ band abstains with zero confidence. The decision rule is $D(x) = \hat{y}$ if confidence $\geq \tau$, else ABSTAIN.

Candidate isolation. Recognition attempts, LLM suggestions, and retrieval references are stored as append-only candidates that reference their evidence and can never overwrite a confirmed value. The LLM adapter (DeepSeek, structured output) is disabled by default; the core pipeline runs end-to-end with the model off.

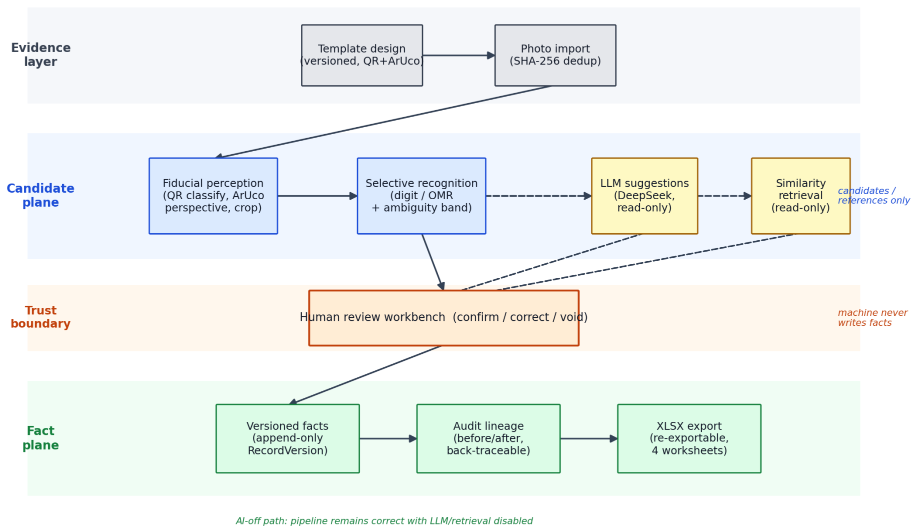

Fig. 1. Trust-constrained architecture of Auto-Decte. The evidence layer feeds a candidate plane (perception, LLM suggestions, retrieval — all candidate/read-only); only the human trust boundary writes versioned facts, with audit lineage and export.

Read-only retrieval. A local, dependency-free similarity index (character-bigram + word cosine) surfaces similar historical forms as references; retrieval results are strictly read-only and never participate in fact modification.

Human decision layer. A React review workbench is the sole path from candidate to fact: operators confirm or correct every value under rule validation; each human action appends a versioned record with a before/after audit pair; exports are back-traceable to form, version, evidence, and event log.

5 System Architecture

Auto-Decte implements the four planes as a single-machine deployment: the evidence layer stores SHA-256-addressed images; the candidate layer holds append-only recognition attempts, isolated suggestions, and read-only retrieval references; the trust boundary is the review workbench; the fact layer holds versioned records, audit events, and export batches with role-based access (operator, reviewer, finance, admin, auditor). A PWA front end with an IndexedDB outbox (scrypt credentials, HttpOnly cookies, CSRF double-submit, idempotency keys) routes in-plant mobile submissions into the same review queue (Fig. 1).

6 Implementation

The backend is FastAPI (Python 3.11) with SQLite via SQLAlchemy 2.0 and Alembic migrations; the front end is React 18 + Vite; image operations use

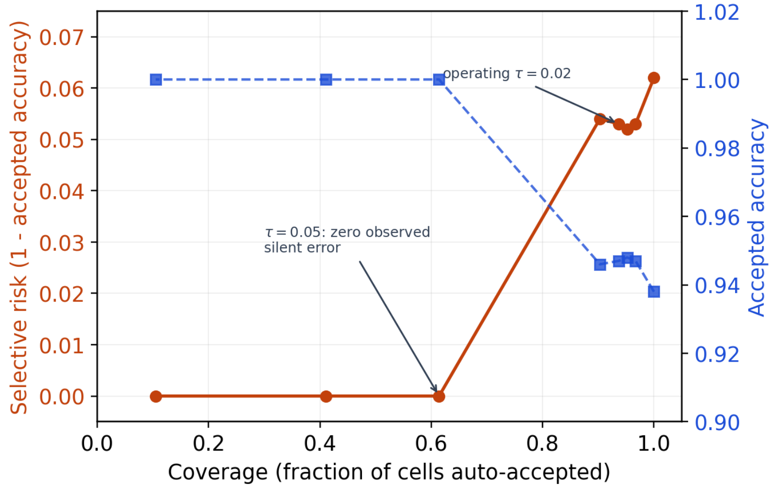

Fig. 2. Coverage vs selective risk over the abstention threshold. Increasing τ lowers coverage but drives the observed selective risk to zero on the benchmark.

OpenCV; suggestions use a DeepSeek client; retrieval is a local vector index. The codebase is covered by 389 passing tests on the public snapshot (159 additional tests require enterprise database fixtures excluded by data policy).

7 Experimental Evaluation

Methodology. Functional acceptance via the automated suite; controlled benchmarks on synthetic filled forms under perturbations (blur, rotation, brightness, Gaussian noise, perspective tilt); a learned baseline and a general-OCR baseline; retrieval precision on a synthetic corpus; and a selective-prediction sweep.

Recognition accuracy. Digit recognition is 93.9% overall across 28 font×perturbation conditions: 100% on the printed template font under blur/rotation/noise/brightness, and 80–90% on unseen fonts with 10–20% of cells abstained — degradation concentrates in the human-routed channel rather than silent errors. OMR is 100% correct outside the ambiguity band and 100% abstained inside it. (The 93.9% aggregate is over single-digit-cell test conditions; the whole-form 97.5% clean figure below is a different metric — cell-level accuracy on 20-cell synthetic forms.)

Selective recognition (abstention). Sweeping the margin threshold τ yields the coverage-vs-risk curve in Fig. 2 and Table 1. At the operating point τ=0.02 the system covers 93.7% of cells at 94.7% accepted accuracy (5.3% selective risk); tightening τ to 0.05 raises accepted accuracy to 100% — eliminating *observed* silent recognition errors on this benchmark — while covering 61.4% and routing the rest to review. This quantifies the intended trade-off: abstention eliminates observed silent errors at the cost of lower coverage.

Table 1. Selective recognition: coverage vs selective risk over the abstention threshold τ.

τ	Coverage	Accepted acc.	Selective risk	Routing rate
0.000 (always guess)	1.000	0.938	0.062	0.000
0.020 (operating)	0.937	0.947	0.053	0.063
0.050	0.614	1.000	0.000	0.386
0.080	0.411	1.000	0.000	0.589

Baseline comparison. A learned HOG + linear-SVM recognizer reaches 100% on the printed font and 79.7–80.3% on unseen fonts — comparable to the template matcher — but has no abstention mechanism, so its errors are silent. Against general OCR (Tesseract, single-character mode), the template matcher wins on the deployment-relevant printed-template regime (100% vs 84–87%), whereas Tesseract generalizes better to unseen fonts (94–95% vs 86–90%), which is outside the printed-form scope; in all cases only the template matcher emits an ambiguity channel.

Template detection and geometry. QR recovery succeeds on 100% of 90 perturbed renders; ArUco correction succeeds on 100% of clean/blur/noise/brightness/tilt images with ≤ 0.34 px alignment error, dropping to 50%/20% at $3°/5°$ rotation.

End-to-end whole-form. Synthetic filled forms (10 digit cells + 10 OMR boxes) through the full pipeline yield cell-level accuracy 97.5% clean and 96.5–98.0% under blur/noise/brightness/tilt, with 4.5–8.5% of digit cells abstained; under $3°$ rotation, correction success drops to 60%. Per-image latency is QR search 2,180 ms, quality 20 ms, correction 35 ms.

Retrieval precision. On a corpus of 200 records across five templates (relevance = same template key), mean precision is 1.00@1, 0.99@3, 0.975@5, 0.95@10.

Trust boundary (fault injection). To verify that machine errors never become facts, we adversarially inject a wrong LLM suggestion (suggested value $\neq$ the confirmed value) into a form with a confirmed record. The confirmed fact is unchanged, the wrong suggestion is stored separately as a candidate, and no record version ever carries the machine value; a subsequent human correction remains the authoritative fact. The acceptance suite additionally enforces the AI-off path (workflow completes with the model disabled), read-only retrieval, and back-traceable exports.

LLM/RAG assistance. Trust is the primary acceptance criterion, so the AI layer is designed for correctness before efficiency. The assistant's efficiency — reviewer time, error-catch rate, suggestion acceptance — is the natural next evaluation step in a planned field pilot (Section 8); it is not claimed here.

8 Industrial Case Study

Auto-Decte is deployed as a working baseline at a bamboo-processing plant, replacing hand-kept paper payroll and production records. Deployment establishes that the system runs in a real industrial setting; the quantitative experiments of Section 6 are, by contrast, conducted on controlled, reproducible benchmarks, because real production data is excluded from the public repository by policy. We therefore do not claim a measured industrial deployment effect: deployment motivates the design and provides the substrate for a future field pilot.

9 Scope and Limitations

This paper addresses *structured printed industrial forms* — the regime the template-guided perception is designed for — not general-purpose document understanding. Within that scope, the following limits remain. General handwritten / Chinese-text / signature OCR is not implemented (only single-cell printed digits and OMR); deployment is single-machine without a completed multi-user capacity test; the evaluation uses controlled synthetic benchmarks rather than a 30–50-form golden set with double entry and real plant images, so accuracy claims are limited to those benchmarks; and the LLM assistant has no measured end-to-end efficiency gain. Future work: general OCR integration, independent task workers, a quantitative human-review-efficiency study (recognition-only vs +retrieval vs +LLM), and a field pilot measuring financial-processing time reduction.

10 Conclusion

Auto-Decte instantiates a trust-constrained human-in-the-loop framework in which machine perception, LLM suggestions, and retrieval produce candidates and references, while humans remain the only source of facts, with full traceability from photograph to exported cell. Selective recognition trades coverage for eliminating observed silent errors on the benchmark, and the AI layer assists without ever writing facts. The system is deployed as a working baseline and provides the substrate for a quantitative field study.

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